

Consider the following problem

$$\min_x \sum_{i \in I} f_i(x_i) \quad s.t. \quad \begin{cases} g_i(x_i) = 0, & i \in \{1, \dots, N\} \\ \sum_{i \in I} A_i x_i = b \end{cases}$$

where $x_i \in \mathbb{R}^{n_x}$, $f_i : \mathbb{R}^{n_x} \rightarrow \mathbb{R}$, $g_i : \mathbb{R}^{n_x} \rightarrow \mathbb{R}^{n_g}$, and $A_i \in \mathbb{R}^{n_A \times n_x}$

Our aim is to exploit the structure of problem to distribute computations on local level as much as possible, as opposed to decomposing the problem,

If we associate multipliers $\lambda_i \in \mathbb{R}^{n_g}$ with g_i and $\mu \in \mathbb{R}^{n_A}$, we will have the following Lagrangian and its derivatives:

$$L(x, \lambda, \mu) = \sum_{i \in I} f_i(x_i) + \lambda_i^T g_i(x_i) + \mu^T (\sum_{i \in I} A_i x_i - b)$$

$$\nabla L(x, \lambda, \mu) = \begin{bmatrix} [\nabla L(x, \lambda, \mu)]_1 \\ \vdots \\ [\nabla L(x, \lambda, \mu)]_N \end{bmatrix}$$

and

$$[\nabla L(x, \lambda, \mu)]_i = \nabla_{x_i} f_i(x_i) + \nabla_{x_i} g_i(x_i)^T \lambda_i + A_i^T \mu$$

where ∇g is the Jacobian of the corresponding function.

Therefore to find the minimizer we solve the following problem;

$$\nabla L(x, \lambda, \mu) = 0 \quad s.t. \quad \begin{cases} g_i(x_i) = 0, & i \in \{1, \dots, N\} \\ \sum_{i \in I} A_i x_i = b \end{cases}$$

Presuming regularity, solving this problem using the Newton method will result in the following system of equations.

$$[KKT] \begin{bmatrix} \Delta x \\ \Delta \lambda \\ \Delta \mu \end{bmatrix} = - \begin{bmatrix} \nabla L(x, \lambda, \mu) \\ \vdots \\ g_i(x_i) \\ \vdots \\ \sum_{i \in I} A_i x_i - b \end{bmatrix} \quad (1)$$

Where Δx is a vector of size $N \times n_x$, $\Delta \lambda$ is of size $N \times n_g$ and $\Delta \mu$ of size n_A with the following KKT matrix;

$$KKT = \begin{bmatrix} \begin{bmatrix} \ddots & \nabla_{x_i}^2 f_i(x_i) + \langle \lambda_i, \nabla_{x_i}^2 g_i(x_i) \rangle & \\ & \ddots & \end{bmatrix} & \begin{bmatrix} \ddots & \nabla_{x_i} g_i(x_i)^T & \\ & \ddots & \end{bmatrix} & \begin{bmatrix} \vdots \\ A_i^T \\ \vdots \end{bmatrix} \\ \begin{bmatrix} \ddots & \nabla_{x_i} g_i(x_i) & \\ & \ddots & \end{bmatrix} & [0] & [0] \\ \begin{bmatrix} \dots & A_i & \dots \end{bmatrix} & [0] & [0] \end{bmatrix}$$

The matrix containing Hessian of the Lagrangian consists of N blocks of size $n_x \times n_x$, block containing the Jacobian of g_i s, has N blocks of $n_x \times n_g$ and the block containing A_i^T has N matrices of size $n_x \times n_A$.

Let us denote $\nabla_{x_i}^2 f_i(x_i) + \langle \lambda_i, \nabla_{x_i}^2 g_i(x_i) \rangle$ with D_i and the matrix containing them $\mathbf{D}$, $\nabla L(x, \lambda, \mu)$ as r_1 , $g(x)$ as r_2 , and $\sum_{i \in I} A_i x_i - b$ as r_3 in the right hand side of 1. Then we can solve for Δx :

$$\begin{aligned} D \Delta x + \nabla g^T \Delta \lambda + A^T \Delta \mu &= -r_1 \\ \Delta x &= -D^{-1} r_1 - D^{-1} \begin{bmatrix} \nabla g^T & A^T \end{bmatrix} \begin{bmatrix} \Delta \lambda \\ \Delta \mu \end{bmatrix} \end{aligned}$$

Where due to the structure of the KKT matrix can be solved for each x_i separately.

$$\Delta x_i = -D_i^{-1} r_{1_i} - D_i^{-1} \begin{bmatrix} \nabla_{x_i} g_i(x_i)^T & A_i^T \end{bmatrix} \begin{bmatrix} \Delta \lambda_i \\ \Delta \mu \end{bmatrix}$$

We will proceed with modifying the right hand side of 1 and the KKT matrix.

$$\begin{aligned} \bar{r}_2 &= r_2 - \nabla g D^{-1} r_1 \\ \bar{r}_3 &= r_3 - A D^{-1} r_1 \\ a &= -\nabla g D^{-1} \nabla g^T \\ b &= -\nabla g D^{-1} A^T \\ c &= -A D^{-1} \nabla g^T \\ d &= -A D^{-1} A^T \end{aligned}$$

where a is block diagonal. The modified system will be:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \Delta \lambda \\ \Delta \mu \end{bmatrix} = - \begin{bmatrix} \bar{r}_2 \\ \bar{r}_3 \end{bmatrix}$$

which can be solved by following:

$$\Delta\mu = (d - ca^{-1}b)^{-1}(ca^{-1}\bar{r}_2 - \bar{r}_3)$$

$$\Delta\lambda = a^{-1}(-\bar{r}_2 - b\Delta\mu)$$

For example consider the following:

$$\min_x \frac{1}{2}x_1^T Q_1 x_1 + \frac{1}{2}x_2^T Q_2 x_2 \quad s.t. \quad \begin{cases} A_1 x_1 = b_1 \\ A_2 x_2 = b_2 \\ C_1 x_1 = C_2 x_2 \end{cases}$$

We will have the following Lagrangian and its gradient.

$$L(x, \lambda_1, \lambda_2, \mu) = \frac{1}{2}x_1^T Q_1 x_1 + \frac{1}{2}x_2^T Q_2 x_2 + \lambda_1^T (A_1 x_1 - b_1) + \lambda_2^T (A_2 x_2 - b_2) + \mu^T (C_1 x_1 - C_2 x_2)$$

$$\nabla L(x, \lambda_1, \lambda_2, \mu) = \begin{bmatrix} Q_1 x_1 + A_1^T \lambda_1 + C_1^T \mu \\ Q_2 x_2 + A_2^T \lambda_2 - C_2^T \mu \end{bmatrix}$$

As these equations are linear, Newton step is not necessary and we have the following system of equations:

$$\begin{bmatrix} Q_1 & 0 & A_1^T & 0 & C_1^T \\ 0 & Q_2 & 0 & A_2^T & -C_2^T \\ A_1 & 0 & 0 & 0 & 0 \\ 0 & A_2 & 0 & 0 & 0 \\ C_1 & -C_2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \\ \mu \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ b_1 \\ b_2 \\ 0 \end{bmatrix}$$

$$x = - \begin{bmatrix} Q_1 & 0 \\ 0 & Q_2 \end{bmatrix}^{-1} \begin{bmatrix} A_1^T & 0 & C_1^T \\ 0 & A_2^T & -C_2^T \end{bmatrix} \begin{bmatrix} \lambda \\ \mu \end{bmatrix} \quad (2)$$

$$\bar{r}_2 = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\bar{r}_3 = 0$$

$$a = \begin{bmatrix} A_1 Q_1^{-1} A_1^T & 0 \\ 0 & A_2 Q_2^{-1} A_2^T \end{bmatrix}$$

$$b = \begin{bmatrix} A_1 Q_1^{-1} C_1^T \\ -A_2 Q_2^{-1} C_2^T \end{bmatrix}$$

$$c = [C_1 Q_1^{-1} A_1^T \quad -C_2 Q_2^{-1} A_2^T]$$

$$d = (C_1 Q_1^{-1} C_1^T + C_2 Q_2^{-1} C_2^T)$$

$$\begin{bmatrix} -A_1Q_1^{-1}A_1^T & 0 & -A_1Q_1^{-1}C_1^T \\ 0 & -A_2Q_2^{-1}A_2^T & A_2Q_2^{-1}C_2^T \\ -C_1Q_1^{-1}A_1^T & C_2Q_2^{-1}A_2^T & -(C_1Q_1^{-1}C_1^T + C_2Q_2^{-1}C_2^T) \end{bmatrix} \begin{bmatrix} \lambda \\ \mu \end{bmatrix} = \begin{bmatrix} \bar{r}_2 \\ \bar{r}_3 \end{bmatrix}$$

$$\mu = -(d - ca^{-1}b)^{-1}(ca^{-1}\bar{r}_2 - \bar{r}_3) \quad (3)$$

$$\lambda = a^{-1}(-\bar{r}_2 - b\Delta\mu) \quad (4)$$

Which is solved by the following algorithm.

Algorithm 1 Basic distributed algorithm for linear quadratic problem

Input: Problem parameters Q_i , A_i , b_i , and C_i

Output: x_i , λ_i , and μ

- 1: $\forall i$: Compute $A_iQ_i^{-1}A_i^T$, $A_iQ_i^{-1}C_i^T$ and send them to λ_i , compute $C_iQ_i^{-1}A_i^T$, $C_iQ_i^{-1}C_i^T$ and send it to μ
 - 2: Compute μ from 3 and send it to λ_i
 - 3: Compute λ_i from 4 and send it to x_i
 - 4: Solve for x_i from 2
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